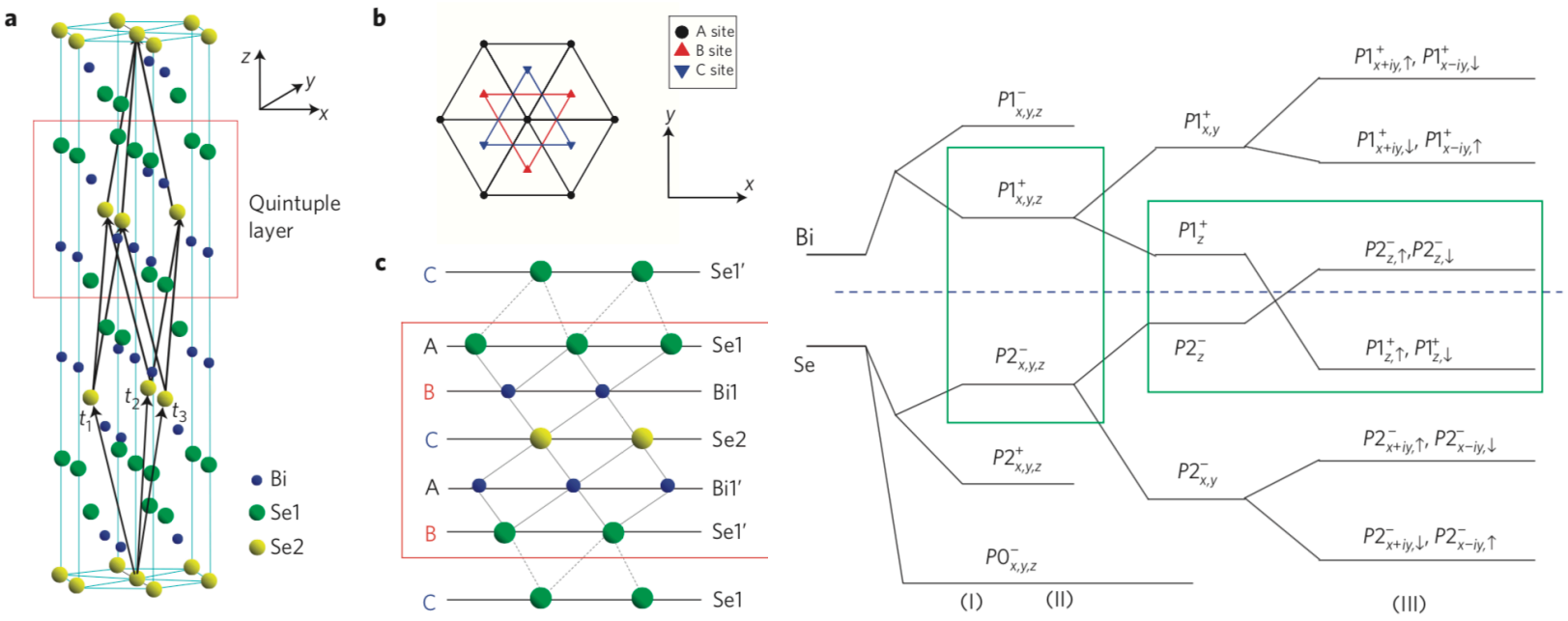
Effective Hamiltonian & finite size effects of TI

Bi_2Se_3[1]



- Can construct a low energy effective four bands model in basis$\{|P1^+_z,\uparrow\rangle, |P2^-_z,\uparrow\rangle, |P1^+_z,\downarrow\rangle, |P2^-_z,\downarrow\rangle\}$

[1] Haijun Zhang, et al, Nature Physics (2009)

Bi_2Se_3

- Symmetries are time reversal Θ, inversion P and 3 fold rotation along z axis R.
- P: inversion does not change the spin, so its representation is $\begin{bmatrix} \tau_z & 0 \\ 0 & \tau_z \end{bmatrix} = I \otimes \tau_z$, where $\tau_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- R: in spin space, rotation of θ is $\exp(i\theta s_z) = \exp(i\frac{\theta}{2}\sigma_z)$, so $R = \begin{bmatrix} e^{i\frac{\theta}{2}} I & 0 \\ 0 & e^{-i\frac{\theta}{2}} I \end{bmatrix} = e^{i\frac{\theta}{2}\sigma_z} \otimes I$

Time reversal symmetry

Acting on state and operator as $|\tilde{\psi}\rangle = \Theta|\psi\rangle, \tilde{H} = \Theta H \Theta^{-1}$

$$i\partial_t|\psi\rangle = H|\psi\rangle \Rightarrow \begin{cases} \tilde{i}\tilde{\partial}_t|\tilde{\psi}\rangle = \tilde{H}|\tilde{\psi}\rangle \\ i\partial_t|\tilde{\psi}\rangle = H|\tilde{\psi}\rangle \end{cases} \Rightarrow \Theta i \Theta^{-1} = -i$$

Θ is an anti-unitary operator: $\langle\tilde{a}|\tilde{b}\rangle = \langle b|a\rangle$, generally, $\Theta = UK$
For spin system:

$$\Theta\sigma_{x,y,z}\Theta^{-1} = -\sigma_{x,y,z} \Rightarrow \begin{cases} U\sigma_{x,z}U^\dagger = -\sigma_{x,z} \\ U\sigma_y U^\dagger = \sigma_y \end{cases}$$

$U = \exp\left(i\frac{\pi}{2}\sigma_y\right) = i\sigma_y,\ \Theta = i\sigma_y K, \Theta^2 = -I$

Kramers pairs

"A time reversal symmetric system with half-integer total spin is at least 2 fold degenerate."
Proof: Since $[H, \Theta] = 0$

- $\Theta|\psi\rangle$ and$|\psi\rangle$ have same energy:
  $H\Theta|\psi_n\rangle = \Theta H|\psi_n\rangle = \Theta E_n|\psi_n\rangle = E_n\Theta|\psi_n\rangle$
- $\Theta|\psi\rangle$ and$|\psi\rangle$ are perpendicular:
  $\langle\Theta\psi|\psi\rangle = \langle\Theta\psi|\Theta^2\psi\rangle = -\langle\Theta\psi|\psi\rangle$

The degenerate states $|\Theta\psi\rangle$ and $|\psi\rangle$ are called Kramers pairs.

Rep. of symmetries

$$\Theta = i\sigma_y \otimes I = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} K, \quad P = I \otimes \tau_z = \begin{bmatrix} \tau_z & 0 \\ 0 & \tau_z \end{bmatrix}$$

$$R = e^{i\frac{\theta}{2}\sigma_z} \otimes I = \begin{bmatrix} e^{i\frac{\theta}{2}} I & 0 \\ 0 & e^{-i\frac{\theta}{2}} I \end{bmatrix}, \theta = \frac{2\pi}{3}$$

Acting on $H(k)$,

$$\Theta H(\boldsymbol{k})\Theta^{-1} = H(-\boldsymbol{k}),\ PH(\boldsymbol{k})P^{-1} = H(-\boldsymbol{k}),$$

$$RH(k_\pm, k_z)R^{-1} = H(e^{\mp i\theta}k_\pm, k_z), \quad k_\pm = k_x \pm ik_y$$

Effective Hamiltonian

Set $H(\boldsymbol{k}) = \begin{bmatrix} A(\boldsymbol{k}) & B(\boldsymbol{k}) \\ B^\dagger(\boldsymbol{k}) & C(\boldsymbol{k}) \end{bmatrix}$, where A and C are Hermitian. Up to second order of k, the relations above can determine the form of A, B, C. For example,

$$A, C(k_\pm, k_z) = A, C(e^{\mp i\theta} k_\pm, k_z),\ e^{i\theta} B(k_\pm, k_z) = B(e^{\mp i\theta}, k_z)$$

require that the leading term of A, C is $k_+ k_- = k_- k_+ = k_x^2 + k_y^2$, while B is k_-. We get the Bernevig-Hugh-Zhang (BHZ) model:

$$H(\boldsymbol{k}) = \epsilon_0(\boldsymbol{k}) I_{4\times 4} + \begin{bmatrix} M(\boldsymbol{k}) & A_1 k_z & 0 & A_2 k_- \\ A_1 k_z & -M(\boldsymbol{k}) & A_2 k_- & 0 \\ 0 & A_2 k_+ & M(\boldsymbol{k}) & -A_1 k_z \\ A_2 k_+ & 0 & -A_1 k_z & -M(\boldsymbol{k}) \end{bmatrix}$$

where $\epsilon_0(\boldsymbol{k}) = C + D_1 k_z^2 + D_2 k_\perp^2, M(\boldsymbol{k}) = M - B_1 k_z^2 - B_2 k_\perp^2$

SnTe

In basis$\{ |A1\uparrow\rangle, |A1\downarrow\rangle, |B1\uparrow\rangle, |B1\downarrow\rangle \}$, the symmetries are:
C_2 rotation: $i\begin{bmatrix}\sigma_z & 0\\ 0 & -\sigma_z\end{bmatrix}$, Mirror reflection M_x: $-i\begin{bmatrix}\sigma_x & 0\\ 0 & -\sigma_x\end{bmatrix}$
M_y:$-i\begin{bmatrix}\sigma_y & 0\\ 0 & \sigma_y\end{bmatrix}$, time reversal T: $i\begin{bmatrix}\sigma_y & 0\\ 0 & \sigma_y\end{bmatrix}K$
By the same procedure, we get the low energy effective Hamiltonian of SnTe:

$$H(\boldsymbol{k}) = \epsilon_0(\boldsymbol{k})I_{4\times4} + \begin{bmatrix} \Delta_0(\boldsymbol{k}) & i\boldsymbol{\alpha}\cdot\boldsymbol{k}_- & i\boldsymbol{\gamma}\cdot\boldsymbol{k}_- & \Delta_1(\boldsymbol{k}) \\ -i\boldsymbol{\alpha}\cdot\boldsymbol{k}_+ & \Delta_0(\boldsymbol{k}) & -\Delta_1(\boldsymbol{k}) & i\boldsymbol{\gamma}\cdot\boldsymbol{k}_+ \\ -i\boldsymbol{\gamma}\cdot\boldsymbol{k}_+ & -\Delta_1(\boldsymbol{k}) & -\Delta_0(\boldsymbol{k}) & i\boldsymbol{\beta}\cdot\boldsymbol{k}_- \\ \Delta_1(\boldsymbol{k}) & -i\boldsymbol{\gamma}\cdot\boldsymbol{k}_- & -i\boldsymbol{\beta}\cdot\boldsymbol{k}_+ & -\Delta_0(\boldsymbol{k}) \end{bmatrix}$$

where $\boldsymbol{\alpha} = (\alpha_x, \alpha_y)$,$\boldsymbol{k}_\pm = (k_x, \pm ik_y)$,$\Delta_0(\boldsymbol{k}) \simeq \Delta_0$,$\Delta_1(\boldsymbol{k}) \simeq \Delta_1$